FOR A SMOOTH PROPER MORPHISM THE RIGHT ADJOINT OF PUSHFORWARD IS THE SHIFT OF THE TOP EXTERIOR POWER OF THE SHEAF OF DIFFERENTIALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0BRT (sources/stacks-duality/pinned-original-source.tex line 3879).

Allowed standard prerequisites (editorial).

- The right adjoint of pushforward and the pre-rigid identity. Same chapter, Lemma labelled lemma-twisted-inverse-image, invoked in the statement, and equation labelled equation-pre-rigid, cited at proof lines 3902-3906. Fixes a and provides the first equality.
- The diagonal of a smooth morphism is a Koszul-regular immersion and hence perfect proper. Divisors, Lemma labelled divisors-lemma-immersion-smooth-into-smooth-regular-immersion, cited at proof lines 3896-3898, and More on Morphisms, Lemma labelled more-morphisms-lemma-regular-immersion-perfect, cited at proof lines 3898-3900. Makes c available and the computation meaningful.
- The comparison with pullback for a flat proper Noetherian morphism and the regular immersion computation. Same chapter, Lemmas labelled lemma-compare-with-pullback-flat-proper-noetherian and lemma-regular-immersion, cited at proof lines 3908-3918. Give the second and fourth equalities of the chain.
- The conormal sheaf of the diagonal is the sheaf of differentials. Morphisms, Lemma labelled morphisms-lemma-differentials-diagonal, cited at proof lines 3928-3930. Identifies C_Delta with Omega_{X/S}, finishing the proof.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line 3894 of the counted proof. It is

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Source reference. Original reference 0B6S: sources/stacks-duality/pinned-original-source.tex, source lines 2945–2945. Same-chapter pointer into the author environment remark at source line 2945; invoked at proof line 3893 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 067U: sources/stacks-chow/tags.txt, source lines 7983–7983. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 3898 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 068C: sources/stacks-chow/tags.txt, source lines 8001–8001. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 3900 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0E2Q: sources/stacks-duality/pinned-original-source.tex, source lines 3166–3166. Same-chapter pointer into the author environment equation at source line 3166; invoked at proof line 3918 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9U: sources/stacks-duality/pinned-original-source.tex, source lines 3306–3306. Same-chapter pointer into the author environment lemma at source line 3306; invoked at proof line 3920 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BR0: sources/stacks-duality/pinned-original-source.tex, source lines 3852–3852. Same-chapter pointer into the author environment lemma at source line 3852; invoked at proof line 3922 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08S2: sources/stacks-chow/tags.txt, source lines 11115–11115. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 3929 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a Noetherian scheme. Let $f : X \to S$ be a smooth proper morphism of relative dimension d. Let a be the right adjoint of $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_S)$ as in Lemma 0A9E. Then there is an isomorphism*

$$\wedge^d \Omega_{X/S}[d] \longrightarrow a(\mathcal{O}_S)$$

in $D(\mathcal{O}_X)$.

Proof. Set $\omega^\bullet_{X/S} = a(\mathcal{O}_S)$ as in Remark 0B6S. Let c be the right adjoint of Lemma 0A9E for $\Delta : X \to X \times_S X$. Because Δ is the diagonal of a smooth morphism it is a Koszul-regular immersion, see Divisors, Lemma 067U. In particular, Δ is a

perfect proper morphism (More on Morphisms, Lemma 068C) and we obtain

$$\begin{aligned}\mathcal{O}_X &= c(L\mathrm{pr}_1^*\omega_{X/S}^\bullet) \\ &= L\Delta^*(L\mathrm{pr}_1^*\omega_{X/S}^\bullet) \otimes^{\mathbf{L}}_{\mathcal{O}_X} c(\mathcal{O}_{X\times_S X}) \\ &= \omega_{X/S}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_X} c(\mathcal{O}_{X\times_S X}) \\ &= \omega_{X/S}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_X} \wedge^d(\mathcal{N}_\Delta)[-d]\end{aligned}$$

The first equality is (0E2Q) because $\omega_{X/S}^\bullet = a(\mathcal{O}_S)$. The second equality by Lemma 0A9U. The third equality because $\mathrm{pr}_1 \circ \Delta = \mathrm{id}_X$. The fourth equality by Lemma 0BR0. Observe that $\wedge^d(\mathcal{N}_\Delta)$ is an invertible $\mathcal{O}_X$-module. Hence $\wedge^d(\mathcal{N}_\Delta)[-d]$ is an invertible object of $D(\mathcal{O}_X)$ and we conclude that $a(\mathcal{O}_S) = \omega_{X/S}^\bullet = \wedge^d(\mathcal{C}_\Delta)[d]$. Since the conormal sheaf $\mathcal{C}_\Delta$ of Δ is $\Omega_{X/S}$ by Morphisms, Lemma 08S2 the proof is complete. □